

Task complexity captures that tasks can vary in their complexity, including how long they take to complete and how difficult they are. A “debugging” task in O*NET could refer to Claude fixing a small error in a function or comprehensively refactoring a codebase—with very different implications for labor demand. We measure complexity through estimated human time to complete tasks without AI, time spent completing tasks with AI, and whether users handle multiple tasks within a single conversation.

Human and AI skills address how automation interacts with skill levels. If AI disproportionately substitutes for tasks requiring less expertise while complementing higher-skilled work, it could be another form of skill-biased technical change—increasing demand for highly skilled workers while displacing lower skilled workers. We measure whether users could have completed tasks without Claude, and the years of education needed to understand both user prompts and Claude’s responses.

Use case distinguishes professional, educational, and personal use. Labor market effects most directly follow from workplace use, while educational use may signal where the future workforce is building AI-complementary skills.

AI autonomy measures the degree to which users delegate decision-making to Claude. Our latest report documented rising “directive” use where users delegate tasks entirely. Tracking autonomy levels—from active collaboration to full delegation—helps forecast the pace of automation.

Task success measures Claude’s assessment of whether Claude completes tasks successfully. Task success helps assess whether tasks can be automated effectively (can a task be automated at all?) and efficiently (how many attempts would it take to automate a task?). That is, task success matters for both the feasibility and the cost of automation labor tasks.

Selecting and validating the new measures

The new dimensions of AI use captured in our data were informed by our recent work on the [productivity effects of Claude](#), feedback we received from external researchers, recent literature on AI’s economic impact through the lens of human capital and expertise ([Vendraminell et al., 2025](#)), and

deliberation within our economic research team. Our main selection criteria were expected economic relevance, complementarity of dimensions, and whether Claude could classify conversations along that dimension with directional accuracy.

We propose that multiple simple primitives, even if somewhat noisy and not perfectly accurate by themselves, can together provide important signals on how AI is being used. We therefore mainly tested for directional accuracy.

For classifying task duration with and without AI, we used minimally modified versions of our [prior productivity work](#). For net new classifiers,¹ implemented via our [privacy-preserving tooling](#), our validation process was as follows. We designed multiple potential measures to capture concepts such as task complexity. For Claude.ai, we evaluated the classifier performance compared to a human researcher on a small set of transcripts in which users gave feedback to Claude.ai and for which we thus have permission to look at underlying transcripts. For first-party API (1P API) data, we validate the classifiers using a mix of internal and synthetic data. Neither data sources are fully representative of Claude.ai or 1P API traffic, but they allow us to check that the classifiers are working on data that resembles real usage data, while ensuring privacy.

Based on initial performance, we revised the classifiers that needed tweaking or discarded classifiers that did not perform well. Interestingly, we find that in some instances (e.g., to measure task success), a simple classifier performed better than a nuanced, complex classifier when compared to human ratings. We then compared performance of classifier versions with vs. without chain of thought prompting, and decided to keep chain of thought prompting only for three facets (human time estimate, human with AI time estimate, and AI autonomy) where we found that it substantially improved performance. We selected a final set of nine new classifiers for the five primitives, all of which are directionally accurate even if they may deviate somewhat from human ratings.

The primitives' value is in what they can predict

Our goal was to create classifiers that are straightforward to implement and in combination provide potentially important economic signals. While we are very confident in the directional accuracy of the new measures (e.g., tasks with higher average years of education needed to understand the human prompt are likely more complex), none of the measures should be taken as exact or definitive (e.g., Claude.ai may somewhat underestimate the human education years needed for many tasks).

Even so, the primitives enrich our understanding of how people use AI. Systematic relationships emerge across primitives, regions, and tasks—patterns we explore in depth in Chapters 3 and 4. That these relationships are intuitive and consistent suggests the primitives capture relevant aspects of how people and businesses use Claude.

External benchmarks reinforce this. In our [productivity work](#), Claude's time estimates correlate with actual time spent on software engineering tasks. Figure 2.1 shows that our human education measure correlates with actual worker education levels across occupations. These validations suggest individual primitives are directionally correct—and combining them may provide additional analytical value, such as enriching productivity estimates with task success rates or constructing new measures of occupational exposure.

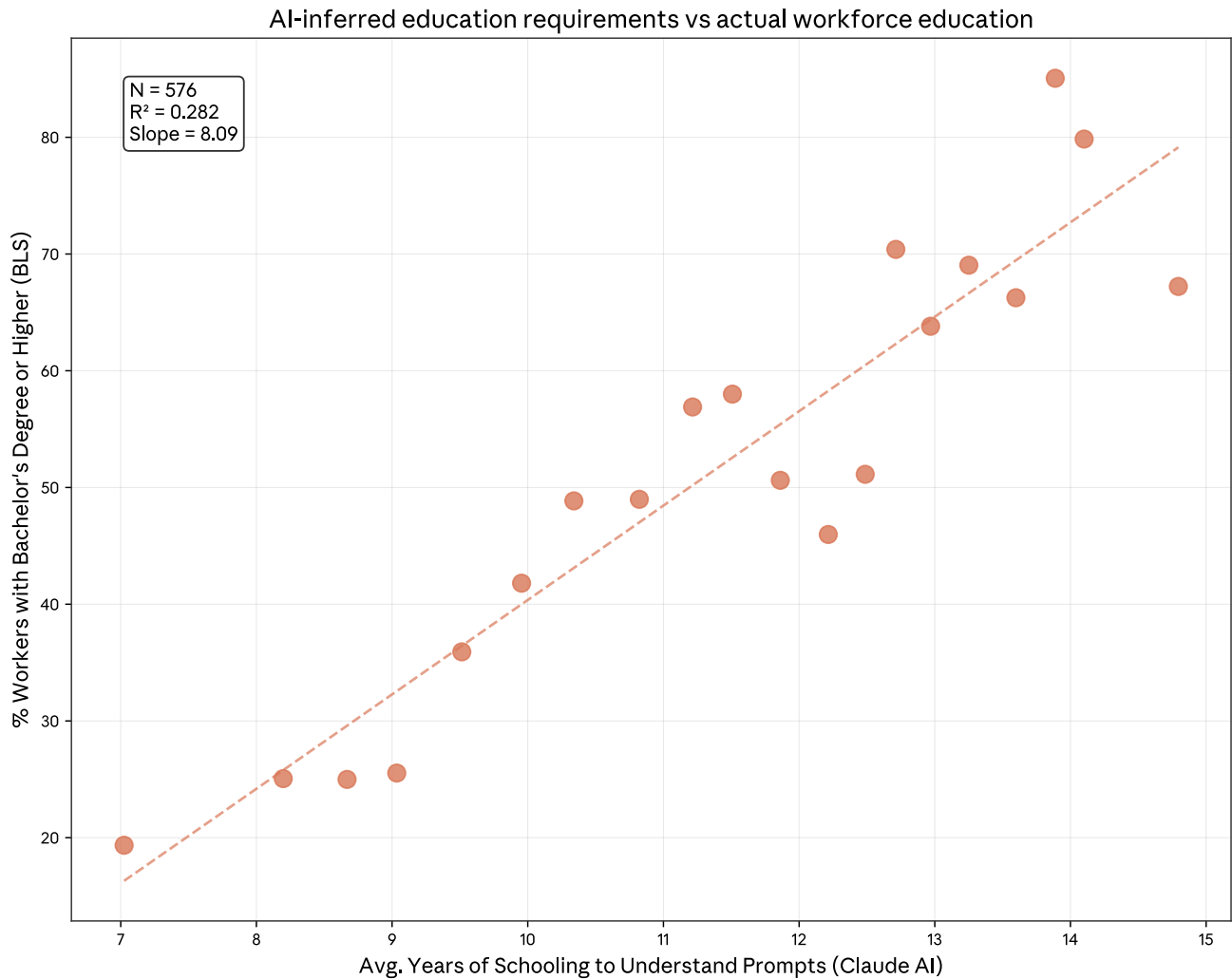


Figure 2.1: Education years needed to understand the human prompt and share of workers with at least a Bachelor's Degree. Education data from "Educational attainment for workers 25 years and older by detailed occupation" (BLS), based on microdata from the 2022 and 2023 American Community Survey.² We calculate average years of schooling for tasks associated with a particular occupation. We then calculate the percentage of workers with a bachelor's degree or higher in that occupation.

Ultimately, the strongest validation will come from the primitives' ability to capture meaningful variation in labor market outcomes. The data we release enable external researchers to analyze economic shifts in new ways. Early work has been encouraging—the automation/augmentation distinction from prior reports has already been used by external researchers to analyze labor market shifts ([Brynjolfsson, Chandar & Chen, 2025](#)).